Spinning - Filament (Melt/Dry/Wet)

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1. Introduction

1.1 Purpose of the study

This document describes the life cycle inventory (LCI) model for synthetic filament yarn production as part of the Carbonfact Textile LCA Database. The model covers melt spinning (POY, FDY, DTY routes), dry spinning (acetate, acrylic, elastane), and wet spinning (MMCF/viscose, polyamide blends), yielding 18 process/material combinations each parameterised across 46 dtex points — 828 rows in total. The model is designed to replace generic filament spinning dataset proxies with LCIs that are scientifically robust, independently verifiable, and harmonised with the staple spinning modules (ring and OE) in the same database.

The central methodological departure of this model from the staple spinning LCIs is the **dtex-independence of specific energy consumption**: unlike ring or OE spinning, filament spinning SEC does not vary with yarn fineness. This is not a simplifying assumption but a physically grounded and empirically validated property of the process, documented in Sections 2 and 3.1 below.

1.2 Spinning process — Melt spinning and filament routes

Synthetic filament yarn is produced by extruding a polymer through a multi-hole spinneret to form continuous filaments. Three solidification mechanisms define the three industrial routes:

Route	Solidification mechanism	Main polymers	Water consumption	Thermal energy
Melt spinning	Quench air cooling (filament solidifies from melt)	PET, PBT, PTT, PLA, PA6, PA66, Acrylic/PAN, Elastane, Acetate, Microfibers, Recycled PA6.6	0.5 kg/kg (quench cooling makeup)	6.20 MJ/kg (melting + extrusion)
Dry spinning	Solvent evaporation in heated chamber	Acetate, Acrylic, Elastane (Spandex)	2.0 kg/kg (condenser cooling)	0 (or 34.65 MJ/kg for acrylic with steam utilities)
Wet spinning	Coagulation in chemical bath	MMCF (Viscose, Lyocell), Polyamide blends	150 kg/kg (coagulation + washing)	0 (ambient/slightly heated bath)

Within melt spinning, three yarn variants are distinguished:

Variant	Description	Electricity (kWh/kg, PET baseline)	Thermal (MJ/kg)	Total (MJ/kg)
POY (Partially Oriented Yarn)	Low orientation; feedstock for draw-texturing or high-speed weaving	0.30	6.20	7.28
FDY (Fully Drawn Yarn)	High orientation; ready for weaving or knitting	0.80	6.20	9.08
DTY (Draw Textured Yarn)	POY + texturing step; texturised for stretch and bulk	0.80 (= 0.30 + 0.50)	6.20	9.08

> **Critical system boundary note:** This model covers **gate-to-gate spinning** (polymer granules to spun yarn). Polymerisation is excluded from the foreground system. For cradle-to-gate analysis, polymer production impacts must be added upstream.

1.3 System boundary

The system boundary covers polymer granules/solution through to finished filament yarn at the spinning mill gate. **Included:** electricity (all routes), thermal energy (melt routes only), lubricant/spin finish input and waste mineral oil output (all routes), process water input and wastewater output (amounts vary by 300× across routes), plastic bobbin input and waste output, capital goods and EoL steel, building infrastructure, and material waste (3% inline — see Section 3.5). **Excluded:** polymerisation and monomer production; downstream texturing beyond DTY increment; dyeing and finishing; packaging and transport.

The yarn loss convention for filament spinning **differs from the staple models**: yarn waste is modelled as an **inline waste output** (3% fixed), not via an external IOR. This means the caller does not need to apply a separate IOR scaling factor. See Section 3.5 for the full explanation.

1.4 Functional unit and scope

Parameter	Value
Functional unit	1 kg of finished filament yarn at the spinning mill gate (gate-to-gate)
System boundary	Polymer granules/solution → spun filament yarn (polymerisation excluded)
Background database	ecoinvent 3.12, system model: cut-off by classification
Background linking	Generic market activities, geography GLO or RoW
Impact method	IPCC 2021 GWP100 via Brightway2
Grid dimensions	18 process/material combinations × 46 dtex points = **828 rows**
dtex	Used as structural metadata only — energy values are **constant across all dtex**
Key departure from staple models	Dtex-independence: SEC does not vary with yarn fineness (Section 3.1)

2. Data sources and literature review

2.1 Primary literature source: Van der Velden et al. (2014)

Van der Velden et al. (2014) provide the most comprehensive measured electricity and thermal energy dataset for filament spinning in the peer-reviewed literature. Their Table 4 reports process-specific SEC from direct plant metering across 12 facilities covering PET, PA6, and acrylic, spanning the dtex range 80–500 dtex. This dataset provides the primary reference for both the process-specific SEC values and the empirical evidence for dtex-independence.

Parameter	Value	Notes
PET FDY melt spinning SEC (electricity)	0.80 kWh/kg	Table 4, mean across 12 plants; used as FDY baseline
PET POY melt spinning SEC (electricity)	0.30 kWh/kg	Table 4; POY requires less drawing energy
Texturing SEC (draw-texturing, DTY)	0.50 kWh/kg	Separate line item, Table 4; added to POY for DTY route
Melt spinning thermal energy	6.20 MJ/kg	Table 4; all melt processes share same extrusion thermal load
dtex range studied	80–500 dtex	PET, PA6, acrylic measured
dtex-dependence finding	**No significant variation**	Confirmed across full 80–500 dtex range; SEC constant within measurement uncertainty
Viscose (MMCF) wet spinning	1.33 kWh/kg	Table 4; no thermal energy (ambient coagulation bath)
Material-specific AII factors	PET=1.00, PA=1.14, PAN=1.17, Elastane=1.34, Acetate=0.80	Table 5; derived from melt viscosity, crystallisation rate, drawing force

2.2 Corroborating evidence for dtex-independence

Beyond the Van der Velden (2014) empirical dataset, dtex-independence for filament spinning is supported by two converging lines of evidence.

First, the physical process analysis (Section 3.1.1) shows that every energy-consuming sub-step in melt spinning — extruder heating, melt pumping, quench cooling, drawing, and winding — is governed by polymer mass throughput per unit time, not by filament geometry. Adjusting for a different dtex target requires changing the spinneret hole configuration, but the total mass flow rate and therefore the total energy per kilogram remain constant. Van der Velden et al. (2014, p. 340) state this explicitly: *"extruding is not a function of decitex, but a function of the extrusion energy of the polymer"*, noting that PET, nylon, and elastane all share the same extrusion energy of 6.2 MJ/kg (citing Cambridge Engineering Selector, CES 2012, as the source for polymer thermal properties).

Second, the acrylic fibre LCA by Yacout et al. (2016), based on plant-level inventory data from a large Egyptian acrylic fibre facility (18,000 t/year), reports an electricity consumption of 1.32 kWh/kg and steam consumption of 9.8 kg steam/kg for wet-spinning acrylic fibre. These values are reported as single process-level constants without any dtex-dependent parameterisation, which is consistent with the mass-governed energy model, although the paper does not explicitly test dtex-independence.

No independent peer-reviewed study has been found that directly tests the hypothesis of dtex-independence for melt spinning across a range of yarn counts using plant-metered data. The empirical evidence base is therefore currently limited to Van der Velden et al. (2014), supported by physical process reasoning and by the absence of any published evidence of a dtex–SEC relationship in the filament spinning literature. If plant-metered data across a range of dtex values becomes available in future, this assumption should be revisited.

2.3 Supporting source: RSC (2023) — acrylic dry spinning

The Royal Society of Chemistry (2023) reports energy data for acrylic dry spinning in their supplementary Table S8, providing the data for the acrylic with steam utilities case: 1.32 kWh/kg electricity and 9.8 kg steam/kg ×

3.54 MJ/kg enthalpy = 34.65 MJ/kg thermal. This case is included in the model as an upper-bound scenario for acrylic dry spinning in facilities that route utility steam through the spinning process.

2.4 The Ecobalyse formula and why it fails for filament spinning

The Ecobalyse parametric model (ADEME, 2024) uses the formula $(\mathrm{Nm}/50) \times C_{\mathrm{filament}}$ for filament spinning electricity, producing an SEC that is proportional to Nm and therefore inversely proportional to dtex. Applied to PET melt spinning, this generates a range of errors that increase systematically with fineness:

Yarn count	dtex	Ecobalyse SEC (kWh/kg)	Measured SEC (kWh/kg)	Error
Nm 25	400	0.75	2.02 (Van der Velden, 2014)	−63% — severe underestimate
Nm 50	200	1.50	2.02	−26% — underestimate
Nm 100	100	3.00	2.02	+49% — overestimate
Nm 200	50	6.00	2.02	+197% — severe overestimate

The Ecobalyse formula is based on the physics of staple spinning (where twist insertion and drafting are count-dependent) and is incorrect for filament spinning (where extrusion, quenching, and drawing are mass-governed processes independent of filament geometry). The Carbonfact model replaces the Ecobalyse formula with dtex-independent SEC values anchored to measured data.

3. Life cycle inventory modelling

3.1 Energy model — the case for dtex-independence

3.1.1 Physical process analysis

Filament spinning SEC is governed by the mass throughput of the spinning line, not by the geometry of individual filaments. The sub-process energy breakdown confirms this at each stage:

Sub-process	Energy share	Dependence on dtex
Extruder heating	~60%	None: depends on polymer mass throughput (kg/h), not filament geometry
Melt pumping	~10%	None: total mass flow governs pump power
Extrusion through spinneret	~5%	None: finer yarn means more holes but same total mass flow and pump power
Quench cooling	~10%	None: mass-based cooling requirement
Drawing	~10%	Approximately none: finer yarn adjusts force but total power per kg is approximately constant
Winding	~5%	None: mass-based winding load

The spinneret geometry adjusts to the target dtex (more holes, smaller diameter), but the total polymer mass flow rate and total energy per kilogram remain constant. This is the fundamental physical reason that filament spinning SEC is dtex-independent.

3.1.2 Empirical validation of dtex-independence

Source	Evidence	Materials	dtex range	Finding
Van der Velden et al. (2014)	Direct plant metering, n=12 facilities	PET, PA6, acrylic	80–500 dtex	No significant variation of SEC with yarn count
Yacout et al. (2016)	Plant-level LCI from Egyptian acrylic facility (18,000 t/yr)	Acrylic (wet/dry spinning)	Not parameterised by dtex	SEC reported as a single process constant (1.32 kWh/kg), consistent with dtex-independence; paper does not explicitly test the hypothesis
Industrial metering data (proprietary, anonymised)	Internal benchmark	Multiple synthetics	100–400 dtex	SEC variability <5% across dtex range, within measurement uncertainty

3.2 Energy model - mathematical formulation

3.2.1 General equations

$$\boxed{E_{\text{elec}}(p,m) = \text{SEC}_{\text{ref,elec}}(p) \times f_{\text{AII}}(m) \times M \quad [\text{kWh}]}$$

$$\boxed{E_{\text{heat}}(p,m) = \text{SEC}_{\text{ref,heat}}(p) \times f_{\text{AII}}(m) \times M \quad [\text{MJ}]}$$

$$E_{\text{total}}(p,m) = (E_{\text{elec}} \times 3.6) + E_{\text{heat}} \quad [\text{MJ}]$$

Variable definitions:

Symbol	Definition	Units
p	Spinning process route: one of {Melt POY, Melt FDY, Melt DTY, Dry, Wet}	—
m	Material family: one of {PET, PA, PAN, Elastane, Acetate, Microfibers, Recycled PA6.6, MMCF, Polyamide blends}	—
M	Mass of finished filament yarn (functional unit), constant at 1 kg	kg
$\text{SEC}_{\text{ref,elec}}(p)$	Reference specific electricity consumption for process route p at the PET baseline $f_{\text{AII}} = 1.00$	kWh/kg
$\text{SEC}_{\text{ref,heat}}(p)$	Reference specific thermal energy consumption for process route p at the PET baseline	MJ/kg
$f_{\text{AII}}(m)$	Dimensionless material adjustment factor for material family m relative to PET $f_{\text{AII,PET}} = 1.00$; sourced from AII/Made2Flow benchmarking dataset (Van der Velden, 2014, Table 5)	—
$E_{\text{elec}}(p,m)$	Electricity consumption for route p and material m, per 1 kg of finished yarn	kWh
$E_{\text{heat}}(p,m)$	Thermal energy consumption for route p and material m, per 1 kg of finished yarn	MJ
$E_{\text{total}}(p,m)$	Total primary energy per 1 kg of finished yarn; electricity converted using 1 kWh = 3.6 MJ	MJ

Dtex-independence: The equations above contain no dtex argument. This is not an omission but a deliberate modelling statement: for filament spinning, SEC is governed by polymer mass throughput, not filament geometry. The value of E_{elec} and E_{heat} is therefore identical across all dtex values in the dataset grid. Dtex appears only as structural metadata labelling each row, with no effect on any calculated quantity. See Section 3.1 for the physical justification and empirical evidence.

3.2.2 Process-specific reference SEC values (baseline: PET = 1.00)

Process	$\text{SEC}_{\text{ref,elec}}$ (kWh/kg)	$\text{SEC}_{\text{ref,heat}}$ (MJ/kg)	E_{total} (MJ/kg)	Source
Melt spinning — POY	0.30	6.20	7.28	Van der Velden (2014), Table 4

Process	$\mathrm{SEC}_{\mathrm{ref,elec}}$ (kWh/kg)	$\mathrm{SEC}_{\mathrm{ref,heat}}$ (MJ/kg)	E_{total} (MJ/kg)	Source
Melt spinning — FDY	0.80	6.20	9.08	Van der Velden (2014), Table 4
Texturing (DTY increment)	0.50	0	1.80	Van der Velden (2014), separate line item
Melt spinning — DTY (= POY + texturing)	0.80	6.20	9.08	Energy-equivalent to FDY
Dry spinning — Acetate	1.00	0	3.60	Estimated; gate-to-gate, no steam utilities
Dry spinning — Acrylic (no steam)	1.20	0	4.32	Estimated
Dry spinning — Acrylic (with steam utilities)	1.32	34.65	39.40	RSC (2023), Table S8
Dry spinning — Elastane	1.50	0	5.40	Estimated
Dry spinning — Generic Extruder Polymer	1.32	0	4.75	Average of acetate, acrylic, elastane
Wet spinning — Viscose (MMCF)	1.33	0	4.79	Van der Velden (2014); BETA (2023)
Wet spinning — Polyamide blends	2.18	0	7.85	Estimated; higher than viscose

> **Note on CED correction for dry spinning:** Van der Velden (2014) reports a CED value of 5.33 kWh/kg for extruder polymer dry spinning. This **cannot** be used directly as process energy because CED includes solvent production energy (~1.5 kWh/kg) and solvent recovery/distillation (~2.5 kWh/kg) — upstream and auxiliary processes outside the gate-to-gate spinning boundary. The gate-to-gate process electricity for generic dry spinning is 1.32 kWh/kg.

3.2.3 Material AII factors

$$\boxed{\mathrm{SEC}_{\mathrm{material}}(p) = f_{\mathrm{AII}}(m) \times \mathrm{SEC}_{\mathrm{ref}}(p)}$$

Material m	f_{AII} (melt)	Physical rationale	Calculated SEC at FDY (kWh/kg)
Polyester (PET, PBT, PTT, PLA)	1.00	Baseline reference material	0.80
Polyamides (PA6, PA66, PA11, PA12)	1.14	Higher melt viscosity (~100 Pa·s vs 50 Pa·s for PET) → higher pump/extruder power; higher drawing force	0.91
Acrylic, Modacrylic, PAN	1.17	High molecular weight (Mw ~100,000 g/mol), slow crystallisation → extended barrel residence time; higher drawing friction	0.94
Elastane (Spandex/Lycra), TPU	1.34	Multi-component extrusion (hard/soft segments, precise temperature control); complex drawing with elastic recovery	1.07
Acetate, Triacetate	0.80	Lower melting point (Tm ~230°C vs 260°C for PET) → reduced heating energy; lower modulus	0.64
Microfibers	1.00	Same composition as baseline PET; spinneret geometry adjusts but total energy per kg unchanged	0.80

Material m	f_{AII} (melt)	Physical rationale	Calculated SEC at FDY (kWh/kg)
Recycled PA 6.6	1.14	Same as virgin PA66; spinning process is unchanged by feedstock origin	0.91

Complete energy table — all melt routes:

Material	POY Elec (kWh/kg)	POY Heat (MJ/kg)	POY Total (MJ/kg)	FDY Elec (kWh/kg)	FDY Heat (MJ/kg)	FDY Total (MJ/kg)
Polyester (PET, PBT, PTT, PLA)	0.300	6.20	7.28	0.800	6.20	9.08
Polyamides (PA6, PA66, PA11, PA12)	0.342	7.07	8.30	0.912	7.07	10.35
Acrylic, Modacrylic, PAN	0.351	7.25	8.51	0.936	7.25	10.62
Elastane (Spandex/Lycra), TPU	0.402	8.31	9.76	1.072	8.31	12.16
Acetate, Triacetate	0.240	4.96	5.82	0.640	4.96	7.26
Microfibers	0.300	6.20	7.28	0.800	6.20	9.08
Recycled PA 6.6	0.342	7.07	8.30	0.912	7.07	10.35

3.3 Thermal energy — melt spinning only

Melt spinning requires thermal energy to heat polymer from ambient temperature to melt temperature and maintain extrusion temperature. The reference value of 6.20 MJ/kg for PET (from Van der Velden, 2014, Table 4) is linked to `market group for heat, district or industrial, natural gas {GLO} | heat, district or industrial, natural gas | cut-off`. This thermal energy is scaled by the AII factor in the same way as electricity. Dry and wet spinning routes have zero thermal energy input (solvent evaporation in dry spinning uses electrical heating of the spinning shaft; wet spinning uses ambient-temperature coagulation).

3.4 Lubricant — spin finish oil

All filament routes apply a spin finish oil immediately after solidification to reduce static, improve winding, and facilitate downstream processing. The spin finish rate is modelled at a **fixed 0.01 kg/kg** (1% of yarn mass), constant across all materials, routes, and dtex values. This value is derived from standard industrial spin finish application rates (0.5–1.5% by mass, mid-range default). The spin finish input is linked to `lubricating oil production {RoW} | lubricating oil | cut-off`; the output waste mineral oil to `market for waste mineral oil {RoW} | waste mineral oil | cut-off`.

3.5 Yarn waste/losses and material input proxy

Filament spinning waste (filament breakage, edge trim, winding losses) is modelled at a **fixed 3% of output mass**, constant across all routes, materials, and dtex values. This is modelled as an **inline waste output** — distinct from the external IOR convention used in the ring and OE staple models:

$$\boxed{\text{Inline waste} = 0.03\ \text{kg/kg finished yarn (all routes, all materials, all dtex)}}$$

The ecoinvent activity for the waste output depends on material:

- Polyester routes: `waste polyester, industrial, from textile production, Recycled Content cut-off {RoW/GLO}`
- All other routes: `market for waste yarn and waste textile {RoW} | cut-off`

> **Important note for database integration:** The inline waste convention for filament spinning means the caller does **not** need to apply a separate IOR scaling factor for filament datasets. The polymer input implicitly required per kg of finished yarn is 1.031 kg (= 1 ÷ (1 − 0.03)). This differs from the ring and OE models and must be documented in any supply-chain integration combining spinning modules from both technologies.

3.6 Process water and wastewater

Process water consumption varies by three orders of magnitude across the three spinning routes, making it one of the key differentiators for environmental assessment:

Route	Water input (kg/kg)	Wastewater output (m^3/kg)	Primary use	Physical basis
Melt spinning	0.5	0.0005	Quench air system cooling (closed-loop with makeup)	Indirect cooling via heat exchanger; minimal direct contact with yarn
Dry spinning	2.0	0.002	Solvent recovery condenser cooling	Higher than melt due to high latent heat of solvent vaporisation
Wet spinning	150	0.15	Coagulation bath (~50 kg/kg) + washing stages (~80 kg/kg) + auxiliary (~20 kg/kg)	300× higher than melt; coagulation requires large bath volume; sequential washing removes residual chemicals

Water input: `market group for tap water {RoW} | tap water | cut-off`. Wastewater output: `market for wastewater, average {RoW} | wastewater, average | cut-off`.

3.7 Capital goods: machinery and buildings

$$Q_{\text{lifetime}} = 15 \text{ yr} \times 8{,}760 \text{ h/yr} \times 0.80 \times \sim 30 \text{ kg/h} \approx 3.16 \times 10^6 \text{ kg}$$

$$(\text{Simplified to } 2{,}155{,}172 \text{ kg for harmonisation with staple models})$$

$$\text{Capital goods: } \boxed{4.64 \times 10^{-7} \text{ unit/kg}} \rightarrow \texttt{market for building machine \{GLO\} | cut-off}$$

$$\text{EoL steel: } 0.0013 \text{ kg/kg} \rightarrow \texttt{market for waste steel \{RoW\} | cut-off}$$

The filament spinning building intensity (**2.52 × 10^{-5} m^2/kg**) is larger than the staple spinning values (ring: 1.0 × 10^{-5}; OE: 8.84 × 10^{-6}) to account for the extrusion zone, quench shaft, and draw-texturing machine footprint.

> **Shared methodology:** Capital Goods & End-of-Life (Machinery) in LCA — Full derivation, lifetime parameters, EoL split. Process-specific value: **4.64 × 10^{-7} unit/kg** for all filament variants, EoL steel **0.0013 kg/kg** — harmonised across all spinning technologies.

> **Shared methodology:** Building Infrastructure in LCA — Process-specific value: **2.52 × 10^{-5} m^2/kg**, constant across all melt/dry/wet variants. Activity: `building, hall {GLO} | building, hall | cut-off`.

3.8 Plastic bobbins

Identical to the staple spinning models: **0.001 kg of polypropylene per kilogram of yarn** → `market for polypropylene, granulate {GLO} | polypropylene, granulate | cut-off`. Output as waste polypropylene assumed to enter recycling: `waste polypropylene, for recycling, unsorted {GLO} | Recycled Content cut-off`. Constant across all routes, materials, and dtex.

3.9 Dataset naming convention

All filament datasets follow: `Spinning to filament, [Material], [Route]`. Example: `Spinning to filament, Polyester (PET, PBT, PTT, PLA etc), Melt Spinning (FDY/DTY)`. Dtex is included as structured metadata in the dataset grid but does not affect energy or mass flows.

4. Life cycle impact assessment

4.1 LCIA results

Climate change impacts (kg CO_2eq / kg yarn) were computed using Brightway2 with the IPCC 2021 GWP100 characterisation factors, applied to the foreground inventory with ecoinvent 3.12 (cut-off) background processes and the GLO average electricity mix. Full results across all 16 EF 3.1 impact indicators are available in `impact-scores.csv`.

Melt spinning — POY vs FDY/DTY comparison (Brightway results)

Process	GHG (kg CO_2eq / kg yarn)
Spinning to filament, Polyester (PET, PBT, PTT, PLA etc), Melt Spinning (POY)	0.541
Spinning to filament, Polyester (PET, PBT, PTT, PLA etc), Melt Spinning (FDY/DTY)	0.885
Spinning to filament, Polyamides (Nylon 6, 6.6, PA 11, PA 12, ...), Melt Spinning (POY)	0.607
Spinning to filament, Polyamides (Nylon 6, 6.6, PA 11, PA 12, ...), Melt Spinning (FDY/DTY)	0.999
Spinning to filament, Microfibers, Melt Spinning (POY)	0.541
Spinning to filament, Microfibers, Melt Spinning (FDY/DTY)	0.885

Melt spinning — material comparison at FDY/DTY

Process	GHG (kg CO_2eq / kg yarn)
Spinning to filament, Acetate, Triacetate, Melt Spinning (FDY/DTY)	0.723
Spinning to filament, Acrylic, Modacrylic, PAN, Melt Spinning (FDY/DTY)	1.02
Spinning to filament, Elastane (Spandex/Lycra), Melt Spinning (FDY/DTY)	1.16
Spinning to filament, Elastane (Spandex/Lycra), Melt Spinning (POY)	0.701

Dry and wet spinning

Process	GHG (kg CO_2eq / kg yarn)
Spinning to filament, Elastane (Spandex/Lycra), Dry Spinning	1.13
Spinning to filament, Extruder Polymer Filaments, Dry Spinning	1.00
Spinning to filament, MMCF (Rayon, Viscose, Lyocell), Wet Spinning	1.08

Results confirm the expected pattern: POY ≈ 0.54–0.61 kg CO_2eq/kg across material families; FDY/DTY ≈ 0.72–1.16 kg CO_2eq/kg depending on material. The All ordering is preserved: Acetate < PET/Microfibers < PA < Acrylic

< Elastane. Values are **dtex-independent** by construction and empirical validation.

4.2 Validation against EF 3.1

Process	Carbonfact GWP (kg CO_2eq/kg)	EF 3.1 comparable	Difference	Interpretation
Melt FDY — Polyester	0.885	1.38 (melt no texturing)	−36%	EF 3.1 grid mix difference; boundary details
Melt FDY — Microfibers	0.885	1.39 (microfibers)	−36%	Same as PET (f_All = 1.00)
Melt FDY — Polyamides	0.999	1.39	−28%	Good agreement
Dry spinning — Extruder Polymer	1.00	1.39 (continuous filament dry)	−28%	Good agreement
Melt POY — Polyester	0.541	1.38 (melt no texturing)	−61%	EF 3.1 likely aggregates POY and FDY; not a model error
Texturing increment	0.50 kWh/kg	0.78 kWh/kg (EF 3.1)	−36%	Within source uncertainty

The systematic downward difference is primarily attributable to the GLO average electricity mix used by Carbonfact versus the European mix implicit in most EF 3.1 datasets. The EF 3.1 values for POY likely aggregate POY and FDY into a single dataset, explaining the large apparent difference at POY level.

5. Inferences and limitations

The dtex-independence of filament spinning SEC is the most important property of this model and is supported by empirical evidence from Van der Velden et al. (2014) and by the physical process analysis in Section 3.1.1. The evidence base is currently limited to a single multi-facility dataset; no independent peer-reviewed study has been found that explicitly tests this hypothesis across a range of dtex. However, this applies to the energy sub-processes; it does not mean that all filament LCI parameters are dtex-independent. Water consumption, for example, could in principle vary with filament count at fixed linear density, but no quantitative evidence for this was found in the reviewed literature. This remains a potential refinement for future model versions.

The All material factors are derived from Van der Velden (2014), Table 5, and supplemented with physical reasoning. The primary uncertainty is for elastane (f = 1.34), which is at the upper end of the range and has fewer independent verification points than PET or PA. The dry spinning SEC values (acetate, acrylic, elastane) carry approximately 25% uncertainty due to limited primary sources and the need for a CED correction to extract gate-to-gate process energy from the Van der Velden CED figures.

Wastewater treatment in wet spinning (viscose) uses a generic `market for wastewater, average` activity that may understate the complexity and environmental burden of viscose-specific chemical baths containing zinc sulphate, sodium sulphate, and carbon disulphide derivatives. This is acknowledged as a limitation and should be revisited when a viscose-specific wastewater treatment dataset becomes available in ecoinvent.

Polymerisation is excluded from the foreground system. Users performing cradle-to-gate analysis must add polymer production impacts upstream of this module. Reference polymer production energy data are documented in the Annex.

6. References

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7. Annex

Glossary

Term	Definition
AII factor	Average Incremental Intensity: material-specific adjustment factor for energy consumption relative to baseline (PET = 1.00)
CED	Cumulative Energy Demand: total primary energy including upstream processes — must **not** be confused with process energy
DTY	Draw Textured Yarn: texturised yarn produced by draw-texturing of POY; energy-equivalent to FDY in this model
FDY	Fully Drawn Yarn: high molecular orientation, ready for weaving/knitting
Gate-to-gate	System boundary: polymer granules/solution to finished yarn (excludes polymerisation)
Cradle-to-gate	System boundary: monomers to polymer to yarn (full upstream including polymerisation)
POY	Partially Oriented Yarn: low molecular orientation; feedstock for DTY or direct use in high-speed weaving
SEC	Specific Energy Consumption: energy per unit mass (kWh/kg or MJ/kg); **dtex-independent** for filament spinning
Spinneret	Die with many small holes through which polymer melt is extruded into filaments
Spin finish	Lubricant/antistatic agent applied to filament immediately after solidification; 0.01 kg/kg in this model

LCI for 1 kg — Melt Spinning FDY/DTY, Polyester (PET, PBT, PTT, PLA), 200 dtex (representative example)

Flow	Value	Unit	Source / Rationale		
Electricity (input)	0.80	kWh	$E_{\text{elec}} = 0.80 \times 1.00 \times 1.0 =$ 0.80 kWh/kg (PET FDY baseline, $f_{text{AII}}=1.00$).	electricity, medium voltage \	cut-off`

Flow	Value	Unit	Source / Rationale		
			`market group for electricity, medium voltage {GLO} \		
Thermal energy (input)	6.20	MJ	$E_{\text{heat}} = 6.20 \times 1.00 \times 1.0 = 6.20$ MJ/kg. `market group for heat, district or industrial, natural gas {GLO} \	heat, district or industrial, natural gas \	cut-off`
Lubricating oil / spin finish (input)	0.01	kg	Fixed spin finish rate: 1% of yarn mass, all routes and dtex. `lubricating oil production {RoW} \	lubricating oil \	cut-off`
Waste mineral oil (output)	0.01	kg	`market for waste mineral oil {RoW} \	waste mineral oil \	cut-off`
Material waste — inline (output)	0.03	kg	3% process waste (filament breakage, edge trim, winding loss). `waste polyester, industrial, from textile production, Recycled Content cut-off {RoW/GLO}`		
Polypropylene bobbin (input)	0.001	kg	Constant across all variants. `market for polypropylene, granulate {GLO} \	polypropylene, granulate \	cut-off`
Waste PP bobbin (output)	0.001	kg	`waste polypropylene, for recycling, unsorted {GLO} \	Recycled Content cut-off`	
Capital goods – machine (input)	4.64×10^{-7}	unit	Harmonised with staple models. `market for building machine {GLO} \	building machine \	cut-off`
EoL machine – steel (output)	0.0013	kg	`market for waste steel {RoW} \	waste steel \	cut-off`
Building infrastructure (input)	2.52×10^{-5}	m^2	Larger than staple spinning. `building, hall {GLO} \	building, hall \	cut-off`
Process water (input)	0.5	kg	Melt spinning quench system cooling makeup. `market group for tap water {RoW} \	tap water \	cut-off`
Wastewater (output)	0.0005	m^3	`market for wastewater, average {RoW} \	wastewater, average \	cut-off`

LCI for 1 kg — Melt Spinning FDY/DTY, Elastane (Spandex/Lycra), TPU

Flow	Value	Unit	Source / Rationale	
Electricity (input)	1.072	kWh	$E_{\text{elec}} = 0.80 \times 1.34 = 1.072$ kWh/kg. `market group for electricity, medium voltage {GLO} \	cut-off`
Thermal energy (input)	8.31	MJ	$E_{\text{heat}} = 6.20 \times 1.34 = 8.308$ MJ/kg. `market group for heat, district or industrial, natural gas {GLO} \	cut-off`
Lubricating oil / spin finish (input)	0.01	kg	`lubricating oil production {RoW} \	cut-off`

Flow	Value	Unit	Source / Rationale	
Waste mineral oil (output)	0.01	kg	`market for waste mineral oil {RoW} \	cut-off`
Material waste — inline (output)	0.03	kg	`market for waste yarn and waste textile {RoW} \	cut-off`
Polypropylene bobbin (input)	0.001	kg	`market for polypropylene, granulate {GLO} \	cut-off`
Waste PP bobbin (output)	0.001	kg	`waste polypropylene, for recycling, unsorted {GLO} \	Recycled Content cut-off`
Capital goods – machine (input)	4.64×10^{-7}	unit	`market for building machine {GLO} \	cut-off`
EoL machine – steel (output)	0.0013	kg	`market for waste steel {RoW} \	cut-off`
Building infrastructure (input)	2.52×10^{-5}	m²	`building, hall {GLO} \	cut-off`
Process water (input)	0.5	kg	Melt route; same as PET. `market group for tap water {RoW} \	cut-off`
Wastewater (output)	0.0005	m³	`market for wastewater, average {RoW} \	cut-off`

LCI for 1 kg — Wet Spinning, MMCF (Rayon, Viscose, Lyocell)

Flow	Value	Unit	Source / Rationale	
Electricity (input)	1.33	kWh	Wet spinning reference SEC (Van der Velden, 2014; BETA 2023); $f_{text{All}} = 1.0$. `market group for electricity, medium voltage {GLO} \	cut-off`
Thermal energy (input)	0	MJ	Zero: ambient/slightly heated coagulation bath.	
Lubricating oil / spin finish (input)	0.01	kg	`lubricating oil production {RoW} \	cut-off`
Waste mineral oil (output)	0.01	kg	`market for waste mineral oil {RoW} \	cut-off`
Material waste — inline (output)	0.03	kg	`market for waste yarn and waste textile {RoW} \	cut-off`
Polypropylene bobbin (input)	0.001	kg	`market for polypropylene, granulate {GLO} \	cut-off`
Waste PP bobbin (output)	0.001	kg	`waste polypropylene, for recycling, unsorted {GLO} \	Recycled Content cut-off`
Capital goods – machine (input)	4.64×10^{-7}	unit	`market for building machine {GLO} \	cut-off`
EoL machine – steel (output)	0.0013	kg	`market for waste steel {RoW} \	cut-off`
Building infrastructure (input)	2.52×10^{-5}	m²	`building, hall {GLO} \	cut-off`

Flow	Value	Unit	Source / Rationale	
Process water (input)	150	kg	Coagulation bath ~50 kg/kg + washing ~80 kg/kg + auxiliary ~20 kg/kg. Source: BETA (2023). `market group for tap water {RoW} \	cut-off`
Wastewater (output)	0.15	m³	Contains dissolved spinning chemicals. `market for wastewater, average {RoW} \	cut-off`. Note: generic treatment may understate complexity of viscose-specific baths.

Complete energy and water parameter table — all 18 process/material combinations

Process / Material	Elec (kWh/kg)	Heat (MJ/kg)	Total (MJ/kg)	Water (kg/kg)	f_{AII}
Melt POY — Polyester	0.300	6.20	7.28	0.5	1.00
Melt POY — Acrylic/PAN	0.351	7.25	8.51	0.5	1.17
Melt POY — Polyamides	0.342	7.07	8.30	0.5	1.14
Melt POY — Elastane	0.402	8.31	9.76	0.5	1.34
Melt POY — Acetate	0.240	4.96	5.82	0.5	0.80
Melt POY — Microfibers	0.300	6.20	7.28	0.5	1.00
Melt POY — Recycled PA6.6	0.342	7.07	8.30	0.5	1.14
Melt FDY/DTY — Polyester	0.800	6.20	9.08	0.5	1.00
Melt FDY/DTY — Acrylic/PAN	0.936	7.25	10.62	0.5	1.17
Melt FDY/DTY — Polyamides	0.912	7.07	10.35	0.5	1.14
Melt FDY/DTY — Elastane	1.072	8.31	12.16	0.5	1.34
Melt FDY/DTY — Acetate	0.640	4.96	7.26	0.5	0.80
Melt FDY/DTY — Microfibers	0.800	6.20	9.08	0.5	1.00
Melt FDY/DTY — Recycled PA6.6	0.912	7.07	10.35	0.5	1.14
Dry — Extruder Polymer Filaments	1.320	0	4.75	2.0	1.00
Dry — Elastane	1.500	0	5.40	2.0	1.00
Wet — MMCF (Viscose, Lyocell)	1.330	0	4.79	150.0	1.00
Wet — Polyamide blends	2.180	0	7.85	150.0	1.00